Math is Explaining and Sharing

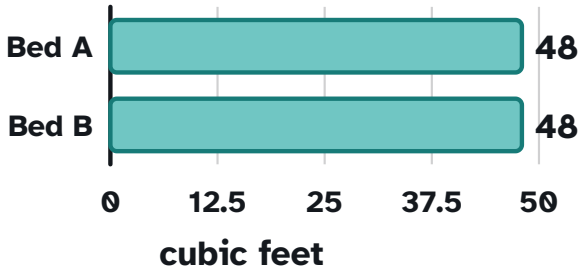
Compare two boxes by volume. If the volumes are equal, give a second reason.

Name _____ **Date** _____

Tonight's spark

A community garden map shows two soil beds. Bed A is 6 ft long, 4 ft wide, and 2 ft deep. Bed B is 8 ft long, 3 ft wide, and 2 ft deep. Which bed holds more soil?

Make a guess first: does Bed A or Bed B hold more soil?



Our class model: a recommendation with two halves

Try it

Work it out on paper. Write one answer on each line.

1 **Which unit fits the volume of a garden bed?**

A. feet **B.** square feet **C.** inches **D.** cubic feet

My answer _____

2 **A bed is 5 ft long, 2 ft wide, and 2 ft deep. What is its volume?**

A. 20 cubic feet **B.** 9 cubic feet **C.** 10 cubic feet **D.** 20 square feet

My answer _____

3 **Two garden beds each hold 24 cubic feet of soil. What can you say?**

A. Bed A holds more **B.** Bed B holds more

C. The volumes are equal, so you need another reason to choose

My answer _____

4 **Bed A is 3 ft by 3 ft by 2 ft. Bed B is 4 ft by 2 ft by 2 ft. Which holds more?**

A. Bed B, because it is longer **B.** Bed A, because 18 cubic feet is more than 16 cubic feet

C. They are equal

My answer _____

5 **A math recommendation has two parts. Which pair is correct?**

A. A guess and a feeling **B.** Only a number **C.** Only an opinion

D. A number you computed and a reason it settles the question

My answer _____

Talk about it

Do the two boxes hold the same amount? What other reason could help you choose?

Home mission

A different situation: use 12 blocks, dice, or ice cubes. Build a box 3 long, 2 wide, and 2 tall. Then build one that is 4 long, 3 wide, and 1 tall. Do they hold the same amount? Which would your student recommend?

If we get stuck, ask

- What number did you find, and what question does it answer?
- Is soil filling a space, covering a surface, or going along a line?
- Can you say your recommendation in one sentence that includes the number?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/1-4-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it